Nhon Trung — Variable Bevel Grinding Test Plan

Date started: _____ **Operator:** _____

CSV file used: `grind_deltas testing with 0.85 deg .5 edge left MODE A.csv`

Blade and setup parameters

Parameter	Value
Stock thickness `T`	1.1 mm
Residual edge `e`	0.5 mm
Bevel height `h`	20 mm
W at heel	35 mm
Bevel mode	A (constant h)
D angle	0.859°
Encoder counts per mm (A, B, C)	1000
Encoder counts per degree (D)	1000
Number of LI segments	200
Execution direction	HEEL → TIP (locked)

How to use this plan

- Go through phases **in order**; do not skip ahead.
- Tick `[x]` each checkbox as you complete it.
- If any step fails, **stop** and note which step caught it at the bottom of this document.
- Run the whole plan on a scrap blade before any real work.

Phase 1 — Bench calibration

(no knife, no grinding)

Tools: dial indicator or feeler gauge, 1.1 mm reference bar (or similar), precision level.

☐ 1.1 Verify fixture alignment (spine truly on pivot)

1. Clamp a straight test bar in the fixture spine-on-pivot as usual.
2. Place dial indicator touching the spine top surface.
3. Rotate D axis through **+2°** and then **−2°**.

PASS: dial reading stays within **±0.02 mm FAIL:** spine moves >0.05 mm → shim the fixture and retry.

Result: ☐ PASS ☐ FAIL Dial swing observed: _____ mm

☐ 1.2 Calibrate `COff` — machine C when stone top = pivot plane

1. Install a 1.1 mm reference bar, spine-on-pivot.
2. Set D = 0 exactly (verify with level).
3. Slowly jog C down until the stone just kisses the TOP of the bar. Paper-drag test: slide paper between stone and bar and stop when you can't pull it through.
4. Read the machine C counter.

COff = _____ **cts** (record here, use throughout plan)

☐ 1.3 Verify C-axis sign direction

1. From COff, jog C by **+100 cts**.
2. Check whether stone goes DOWN (into the bar) or UP (away from bar).

PASS: stone moves into the bar. **FAIL:** stone moves away → C axis sign is flipped; negate all dC values in DMC program.

Result: ☐ PASS ☐ FAIL (flip needed)

☐ 1.4 Verify D-axis sign direction

1. Back C off by 10 mm; remove bar.
2. Jog D by **+500 cts** (= +0.5°).
3. Place level on the clamp; check edge tilt direction.

PASS: edge tilts **UP** (edge side of clamp rises). **FAIL:** edge tilts down → negate all dD values.

Result: ☐ PASS ☐ FAIL (flip needed)

☐ 1.5 Verify A-axis sign direction for heel→tip

1. Clamp the Nhon Trung blade normally (heel end in fixture).
2. Jog A by **−1000 cts**.
3. Watch which direction the grinding zone of the stone moves along the blade.

PASS: stone moves toward TIP of the blade. **FAIL:** moves toward heel → negate dA column.

Result: ☐ PASS ☐ FAIL (flip needed)

Phase 2 — Dry run

(stone backed off, no contact)

☐ 2.1 Choose grinding side

Inspect fixture: does edge face **LEFT** or **RIGHT** on the machine?

Using column: ☐ `dBL_heel_cts` (edge faces left) ☐ `dBR_heel_cts` (edge faces right)

☐ 2.2 Verify B-axis sign

1. Back C off 10 mm.
2. From a mid-position, jog B by **+100 cts**.
3. Observe motion.

For **LEFT**-facing edge (dBL column): stone should move toward the edge side (into knife body on left). For **RIGHT**-facing edge (dBR column): check sign carefully — dBR values are negative at heel→belly, so +100 on the BR axis should match the machine's actual direction convention.

If mismatched — negate that whole column before loading.

Result: ☐ sign matches ☐ need to negate column

☐ 2.3 Pre-position (heel start, NO contact)

Execute on machine:

```
PA A = 147253          ; ArawHeel_cts row 1
PA B = 16342           ; BrawHeel_cts row 1  (or -16342 for BR)
PA D = 859             ; 0.859° × cpd_d
BG A, B, D
AM A, B, D
```

PASS: heel aligns with stone, D tilted 0.859°, C still parked high (no contact).

Result: ☐ PASS ☐ FAIL

☐ 2.4 Load first 10 LI segments only

Load rows 2–11 of CSV into LI buffer (10 segments, keep stone 10 mm backed off).

Expected totals over the 10 segments:

- A moves ≈ **−7,370 cts**
- B moves ≈ **+590 cts** (or −590 for BR)
- C moves ≈ **+10 cts** (barely)

Execute: `BG S; AM S`

PASS: smooth coordinated motion. **FAIL:** buffer error or axis lag → resolve before full buffer.

Result: ☐ PASS ☐ FAIL

☐ 2.5 Reset and load all 200 segments

Return to heel start (as in 2.3). Load all 200 LI segments. Stone still backed off.

Expected end state (after buffer completes):

- A ≈ **0 cts**
- B ≈ **0 cts**
- C ≈ (start) − 245 cts (full C travel across blade)

PASS: smooth traverse heel→tip, final position as expected.

Result: ☐ PASS ☐ FAIL

Phase 3 — First contact (empirical startPtC find)

This is the most important measurement. Find the **actual** machine C for grind contact at the heel.

☐ 3.1 Park at heel position

Use A, B, D from step 2.3.

☐ 3.2 Measure actual contact depth

1. Use dial indicator (or feeler gauge).
2. Lower stone slowly.
3. Stop when dial reads **0.5 mm** clearance between stone and blade edge (= your intended residue e).

C_contact_measured = _____ cts

☐ 3.3 Compare measured to formula-predicted

Formula:

```
C_grind_predicted = COff - 475
```

C_grind_predicted = _____ cts (= COff from step 1.2 minus 475)

Difference = C_contact_measured − C_grind_predicted = _____ cts

PASS: difference within **±25 cts** (±25 μm) — formula matches reality. **FAIL:** difference > 50 cts — investigate:

- Measure T with calipers at 3+ points. T = _____ mm

- Re-verify spine on pivot (repeat 1.1). Spine swing: _____ mm
- Check e interpretation

Result: ☐ PASS ☐ FAIL

☐ 3.4 Record operational values for this blade

- **startGrindC** (use for this grind): `C_contact_measured` = _____ cts
- **restC** (10 mm above grind): `startGrindC - 10000` = _____ cts

Phase 4 — Short test grind

(stone contacting, first 30 segments only)

☐ 4.1 Prepare shortened DMC program

Use ONLY rows 2–31 of CSV (30 segments = ~22 mm heel region). This grinds only the heel third, as a safety test.

☐ 4.2 Execute short grind

Program structure:

```
PA A=147253, B=±16342, D=859            ; pre-position
PA C = restC                            ; back C off
BG A, B, D, C; AM

                                        ; operator confirms position

PA C = startGrindC                      ; descend to grind
BG C; AM C

LS                                      ; run 30 LI segments
...
LE
BG S; AM S

PA C = restC                            ; retract to rest
BG C; AM C
```

Observations:

PASS: smooth traverse, visible grind mark only on heel third. **FAIL:** mark walks off blade / sparks wrong / no contact → stop and diagnose.

Result: ☐ PASS ☐ FAIL

☐ 4.3 Measure residual edge thickness at heel region

Three caliper readings along the ground heel section:

- Point 1 (near heel): e = _____ mm
- Point 2 (middle): e = _____ mm
- Point 3 (end of region): e = _____ mm

Target: **0.5 mm** Tolerance: **±0.05 mm**

PASS: all three readings in 0.45–0.55 mm.

FAIL analysis:

- All three THICKER than target → stone didn't reach deep enough → add (shortfall × cpm_c) to C descent.
- All three THINNER than target → stone went too deep → reduce C by overshoot × cpm_c.
- MIX of thicker and thinner → T isn't constant (measure blade thickness) or spine isn't perfectly on pivot.

Result: ☐ PASS ☐ FAIL Diagnosis: _____

☐ 4.4 Measure bevel height

Bevel width (edge to top of ground area), use calipers: h_measured = _____ mm

Target: **20 mm** Tolerance: **±1 mm**

PASS: 19–21 mm. **FAIL:**

- Shorter than 20 mm → D angle is steeper than specified (check D sign and cpd_d).
- Longer than 20 mm → D angle is shallower.

Result: ☐ PASS ☐ FAIL

Phase 5 — Full grind

(all 200 segments)

Only proceed if Phase 4 passed.

☐ 5.1 Run the full 200-segment program

Same structure as 4.2, but load all 200 segments.

Result: ☐ PASS ☐ FAIL (describe) _____

☐ 5.2 Residual thickness at 5 points along the blade

Location	e measured (mm)	Pass (0.45–0.55)?
Heel	_____	☐
Quarter	_____	☐
Middle	_____	☐
Three-quarter	_____	☐
Near-tip	_____	☐

Drift pattern analysis:

- e INCREASES toward tip → C compensation under-tracking (check cpm_c or W formula).
- e DECREASES toward tip → C compensation over-tracking.
- e OSCILLATES → polyline artifact (kink at specific row — recall rows 42–44).

Pattern observed: _____

☐ 5.3 Bevel height at same 5 points

Location	h measured (mm)	Pass (19–21)?
Heel	_____	☐
Quarter	_____	☐
Middle	_____	☐
Three-quarter	_____	☐
Near-tip	_____	☐

Near-tip may narrow — expected when W approaches h. Document numbers.

☐ 5.4 Visual inspection

- Bevel continuous heel to ~80% of blade length? ☐ Yes ☐ No
- Near-tip region disappearing bevel (expected)? ☐ Yes ☐ No
- Any surface defects (chatter, burn marks, skipped areas)? ☐ None ☐ Some: _____

Phase 6 — Summary

Measurement	Target	Actual	Pass?
COff	—	_____ cts	—
C_contact_measured	COff − 475	_____ cts	☐
Diff (measured − predicted)	≤ ±25 cts	_____ cts	☐

Measurement	Target	Actual	Pass?
e at heel	0.5 ± 0.05	_____ mm	☐
e at middle	0.5 ± 0.05	_____ mm	☐
e at near-tip	0.5 ± 0.05	_____ mm	☐
h at heel	20 ± 1	_____ mm	☐
h at middle	20 ± 1	_____ mm	☐
h at near-tip	20 ± 1 (may narrow)	_____ mm	☐
Bevel continuous ≥ 80%	yes	☐ yes ☐ no	☐

Failures encountered (if any)

Phase / Step that failed: _____

Description: _____

What did you try to fix it? _____

Sign flips you had to apply (if any)

Axis	Original sign	Flipped?	Notes
A	dA_heel_cts	☐ yes ☐ no	
B	dBL or dBR	☐ yes ☐ no	
C	dC_heel_cts	☐ yes ☐ no	
D	dD_heel_cts	☐ yes ☐ no	

Key diagnostic number to report back

C_contact_measured − C_grind_predicted = _____ cts

This single number tells us:

- Close to 0 → formula is geometrically correct on your fixture.
- Positive (say, +2000 cts = 2 mm deeper than predicted) → fixture pivot is not exactly at spine top, OR T is off.
- Negative → stone retracts further than predicted → similar causes, opposite direction.

Decision after testing

Based on this test: ☐ Formula matches, proceed with production. ☐ Offset needed → add to `C0ff` calibration: _____ cts. ☐ Sign flips needed (see above). ☐ Geometry doesn't match — schedule deeper investigation.

Date completed: _____ Operator signature: _____

Notes page overleaf

Notes